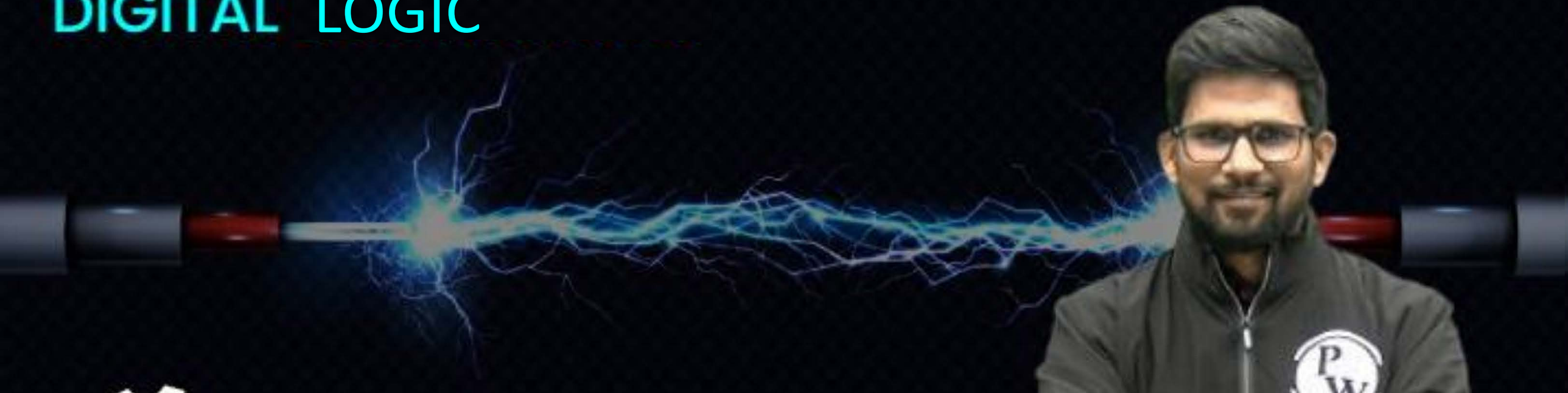
COMPUTER SCIENCE & IT

DIGITAL LOGIC





Lecture No. 01

BOOLEAN THEOREMS AND GATES



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- Theorems & GATES

Decimal no. system:

Base ⟶ (10)

↳ (0 − 9) digits

10^2	10^1	10^0
7	8	6

$= (7 \times 10^2 + 10^1 \times 8 + 10^0 \times 6)$

$(1\ 0)$ ⟶ is a number in decimal number system

$10^1\ 10^0$

$(1\ 9)$

$= 1 \times 10 + 9 \times 10^0$

Binary no. system:

Base → 2
↳ (0 – 1)

$$\overset{2^2\,2^1\,2^0}{(1\,1\,0)_2} = (2^2\times1 + 2^1\times1 + 2^0\times0)_{10} = (6)_{10}$$

$(121)_2$ ⟶ invalid no.

- $\overset{2^3\,2^2\,2^1\,2^0}{(1\,0\,1\,1)_2} = (2^3\times1 + 2^2\times0 + 2^1\times1 + 2^0\times1)_{10} = (11)_{10}$

$\overset{2^3\,2^2\,2^1\,2^0}{(1\,1\,0\,1)_2} = (13)_{10}$

$(1\,0\,1\,0\,1)_2 = (21)_{10}$

MSB → Most significant bit

LSB → Least significant bit

$(1\,1\,0\,1\,1\,0)_2 = (54)_{10}$

- $(23)_{10} = (10111)_2 = (\overset{2^4\ 2^3\ 2^2\ 2^1\ 2^0}{1\ 0\ 1\ 1\ 1})_2$

$(0000)_2 = (0)_{10}$

$\vdots \qquad \vdots$

$(1111)_2 = (15)_{10}$

$(37)_{10} = (\overset{2^5\ 2^4\ 2^3\ 2^2\ 2^1\ 2^0}{1\ 0\ 0\ 1\ 0\ 1})_2$

$(47)_{10} = (101111)_2$

$(999)_{10} = (10^3 - 1)$

10^2	10^1	10^0
9	9	9

$(9999)_{10} = (10^4 - 1)$ $\quad (111\cdots n \text{ times}) = (2^n - 1)_{10}$

$(111)_2 = (7)_{10} = (2^3 - 1)_{10}$

$(1111)_2 = (15)_{10} = (2^4 - 1)_{10}$

Octal no system $\rightarrow$ base $\rightarrow 8 \rightarrow (0-7)$

$(\overset{8^2\ 8^1\ 8^0}{777})_8 = (8^2 \times 7 + 8^1 \times 7 + 8^0 \times 7)_{10}$

$= (511)_{10} = (8^3 - 1)_{10}$

$$(87)_{10} = \begin{matrix} 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 0 & 1 & 0 & 1 & 1 & 1 \end{matrix}_2$$

$$(189)_{10} = \begin{matrix} 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 0 & 1 & 1 & 1 & 1 & 0 & 1 \end{matrix}_2$$

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H.W.

$(29)_{10} = (\quad)_2$

$(111)_{10} = (\quad)_2$

$(150)_{10} = (\quad)_2$

$(199)_{10} = (\quad)_2$



2^1	2^0	
A	B	
0	0	$\rightarrow (0)$
0	1	$\rightarrow (1)_{10}$
1	0	$\rightarrow (2)_{10}$
1	1	$\rightarrow (3)_{10}$

→ Total 4 Combinations

2^2	2^1	2^0	
A	B	C	
0	0	0	$\rightarrow (0)_{10}$
0	0	1	$\rightarrow (1)_{10}$
0	1	0	$\rightarrow (2)_{10}$
0	1	1	$\rightarrow (3)_{10}$
1	0	0	$\rightarrow (4)_{10}$
1	0	1	$\rightarrow (5)_{10}$
1	1	0	$\rightarrow (6)_{10}$
1	1	1	$\rightarrow (7)_{10}$

→ 8 Combinations

A	B	C	D
0	0	0	0 → 0
0	0	0	1 → 1
0	0	1	0 → 2
0	0	1	1 → 3
0	1	0	0 → 4
0	1	0	1 → 5
0	1	1	0 → 6
0	1	1	1 → 7
1	0	0	0 → 8
1	0	0	1 → 9
1	0	1	0 → 10
1	0	1	1 → 11
1	1	0	0 → 12
1	1	0	1 → 13

A	B	C	D
1	1	1	0 → 14
1	1	1	1 → 15

n-variables ⟶ 2^n combinations (terms)

↳ $(0)_{10} \longrightarrow (2^n - 1)_{10}$

2^1	2^0	
A	B	
0	0	⟶ 0
0	1	⟶ 1
1	0	⟶ 2
1	1	⟶ 3

2^2	2^1	2^0	
A	B	C	
0	0	0	⟶ (0)
0	0	1	⟶ (1)
0	1	0	⟶ (2)
0	1	1	⟶ (3)
1	0	0	⟶ (4)
1	0	1	⟶ (5)
1	1	0	⟶ (6)
1	1	1	⟶ (7)



2 Minute Summary

Basics of no system

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Thank you GW Soldiers !

